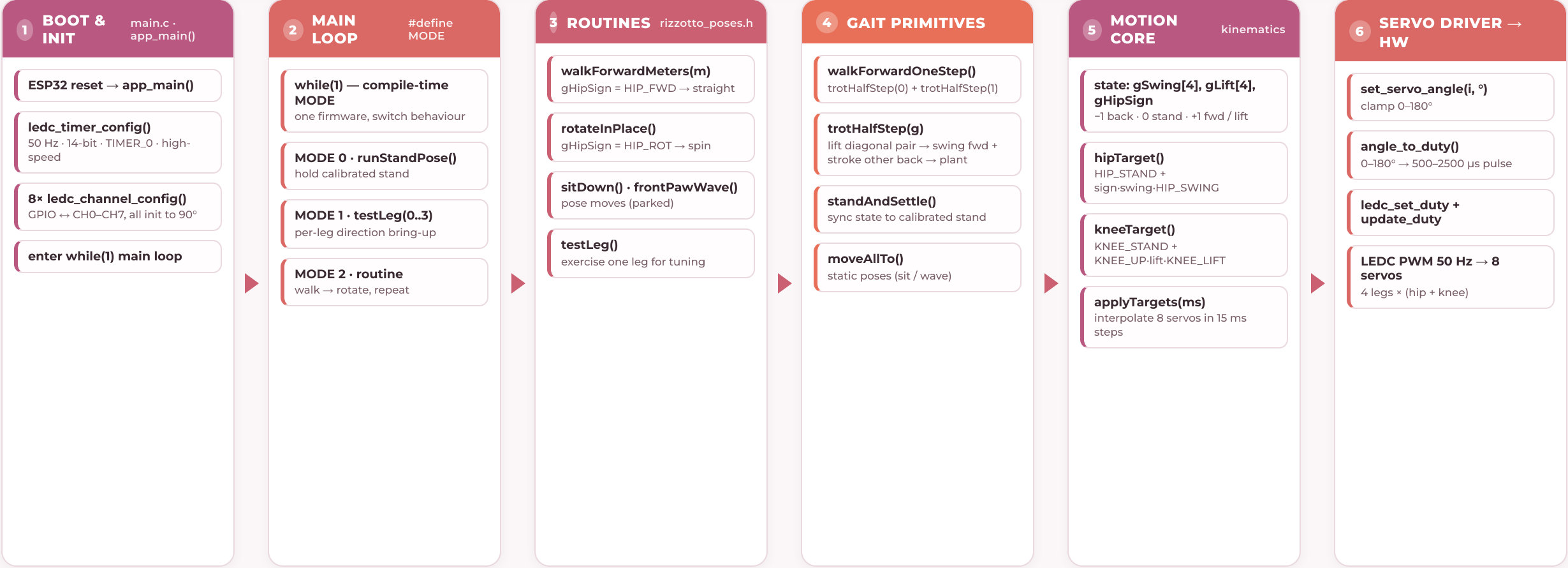


RizzOtto · Quadruped Walking Firmware — Architecture

ESP32 · ESP-IDF (C) · 8× micro-servos (4 legs × hip+knee)
Pipeline: boot → mode → routine → gait → kinematics → PWM → servos



CALIBRATION & TUNING

Per-servo data feeding the Motion Core ▲ — set on hardware

HIP_IDX / KNEE_IDX
leg → physical servo map

HIP_FWD {-1,-1,+1,+1}
straight: front opposite back

HIP_SWING 26 · KNEE_LIFT 36
stride length · foot clearance (°)

HIP_STAND / KNEE_STAND
neutral “stand” angles per servo

HIP_ROT {+1,+1,+1,+1}
spin: all hips same sense

SWING_MS 110 · LIFT_MS 45
move timing — smaller = faster

DIAG pairs
{FL,BR} & {FR,BL} trot diagonals

KNEE_UP
per-leg lift direction signs

CYCLES_PER_METER 14 · TURN_CYCLES 8
distance · rotation amount